SafeContain AI

Smarter Threat Containment. Less Business Disruption.

Maps connected risks and simulates responses to guide better decisions

Reveals attack paths and shows routes that remain after action

Considers business impact to preserve critical workflows

Creates tamper-evident records of decisions and approvals



INTERACTIVE PROTOTYPE & REPOSITORY:
https://github.com/saanviinegii/SafeContain-AI

The Problem

A Blocked Attack Path Is Not Always a Contained Threat.

Fragmented visibility

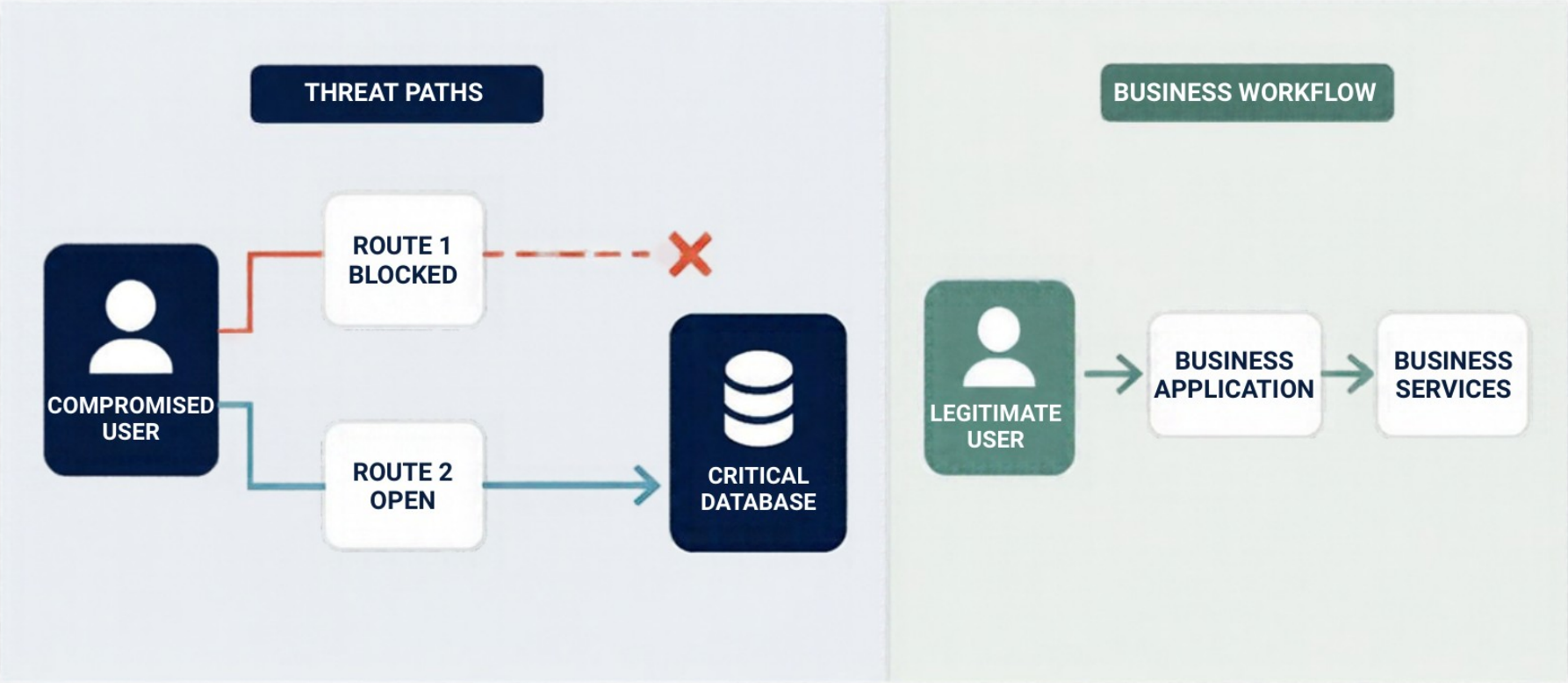
Alerts, accounts, devices, permissions, and assets are seen as disconnected.

Incomplete containment

Blocking one attack path may leave alternative routes open.

Operational disruption

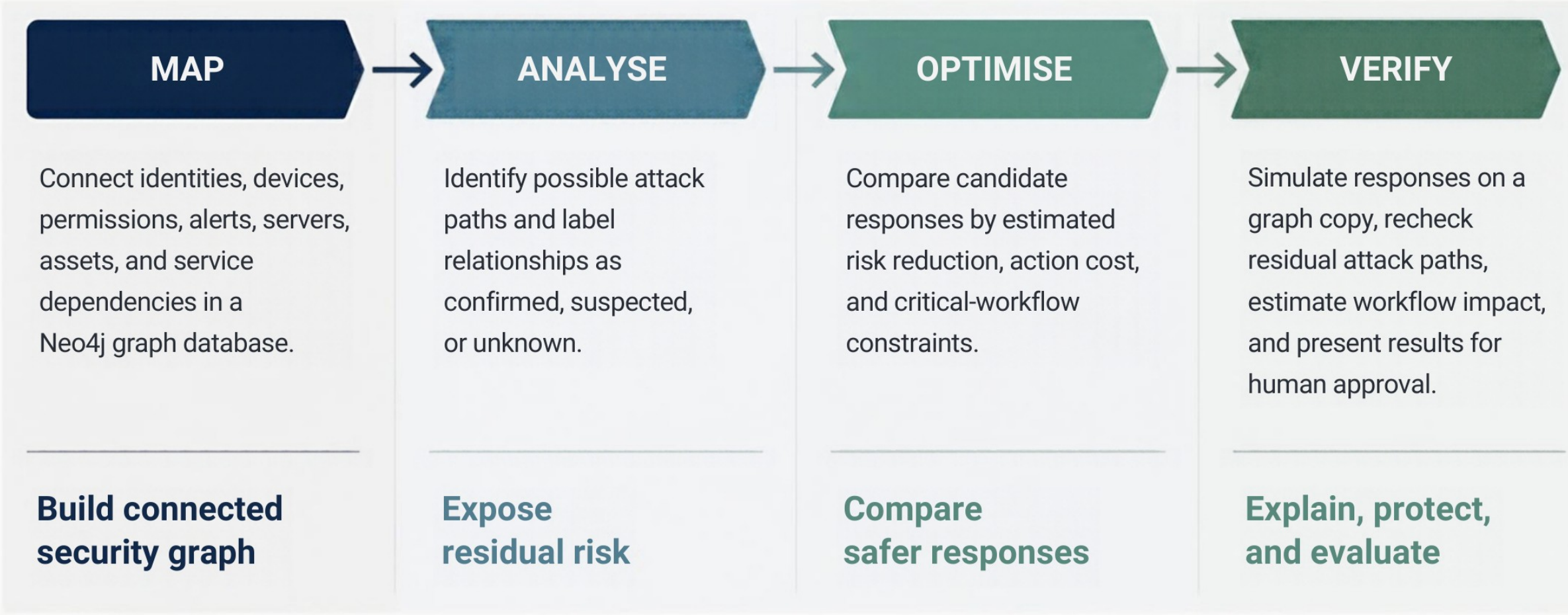
Disabling shared accounts or services risks interrupting legitimate workflows.



The challenge is to reduce security risk without unnecessarily disrupting essential services.

Our Proposed Solution

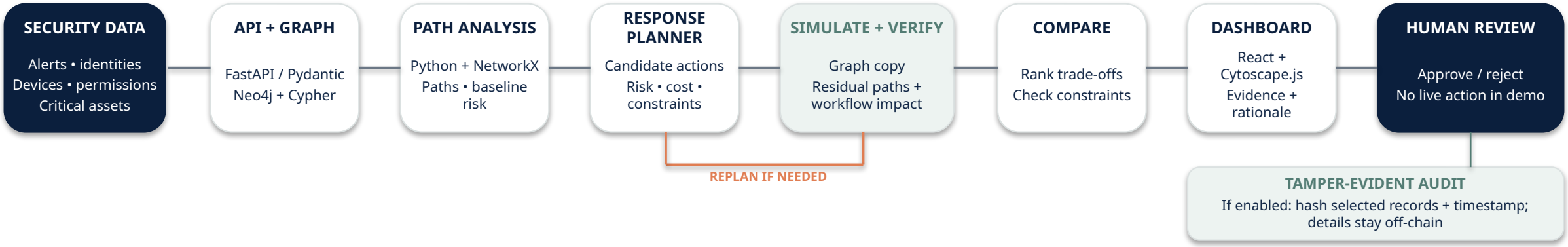
Map. Analyse. Optimise. Verify.



How SafeContain AI Works

PROPOSED ARCHITECTURE

Connected Data. Verified Decisions.



Technology Stack

GRAPH DB	BACKEND	GRAPH ANALYSIS	PLANNING	FRONTEND
Neo4j Cypher	Python FastAPI • Pydantic	NetworkX Path checks	Greedy baseline Optional OR-Tools	React • TypeScript Cytoscape.js

AUDIT	STORAGE	TESTING	DEPLOYMENT	EVALUATION
Optional blockchain (test ledger)	Controlled off-chain evidence	pytest • synthetic scenarios	Docker • optional hosting	Risk • cost • workflow • runtime

Accuracy Notes

- Risk and workflow impact are estimates based on modelled data.
- Confirmed / suspected / unknown are evidence-status labels.
- Recheck residual paths; full containment is not guaranteed.
- Optional blockchain detects record changes, not evidence truth.
- Demo actions are simulated; human approval is separate.

Research Gap and Validation

From Suggested Actions to Measurable Outcomes.

EXISTING CHALLENGE	SAFE CONTAIN AI APPROACH	INTENDED CONTRIBUTION
Fragmented security relationships	Neo4j graph modelling	Clearer visibility of attack paths
Alternative routes remain open	Simulate and recheck paths	Identify residual attack paths
Response actions disrupt services	Workflow-constrained planning	Evaluate security-continuity trade-offs
Incomplete or uncertain evidence	Confirmed, suspected, and unknown relationships	Make assumptions explicit
Decisions are hard to audit	Hash + timestamp records	Detect changes to selected records

EVALUATION PLAN

Use the same reproducible scenarios for each baseline.

1. Severity-first remediation
2. Greedy risk-reduction per cost
3. Exact optimiser on small graphs (if implemented)

Measure residual weighted risk, workflows preserved, action cost, constraint violations and runtime. Report measured results only.

LIMITATIONS

- Input data quality and completeness
- Simulated graphs cannot capture all dependencies
- Uncertain relationships may be misclassified
- Blockchain does not validate evidence truth
- Prototype cannot guarantee attack prevention

RESEARCH GAP

Focused, measurable evaluation of residual path verification, uncertainty representation, workflow constraints, and tamper-evident auditing combined.

Existing work addresses related areas, novelty requires focused literature and product comparison; not proven here.

Demonstration and Expected Impact

Test the Response Before You Trust It.

ILLUSTRATIVE SCENARIO
No measured results claimed

SCENARIO COMPARISON

Suspicious account with two attack routes to critical database

OPTION A

Disable shared service

may disrupt workflows

- Attack paths blocked
- Alternative paths remaining
- Workflow disruption
- Response cost
- Residual risk

OPTION B

Targeted permission change

supported by evidence

- Attack paths blocked
- Alternative paths remaining
- Workflow disruption
- Response cost
- Residual risk

RESPONSE WORKFLOW

Detect → Map → Plan → Simulate → Verify → Compare → Human review → Audit

Verify remaining attack paths and estimated impact on modelled critical workflows before human review.

SIMULATE

Compare outcomes across effectiveness, workflows, cost, and residual risk.

HUMAN REVIEW

Reviewer examines evidence and approves or rejects the recommended plan.

AUDIT INTEGRITY

If enabled, selected records are hashed and timestamped.

DEMONSTRATION ROADMAP

01 Create synthetic/authorized test data; build Neo4j graph

02 Implement path analysis and residual path verification

03 Develop response scoring and simulation

04 Build dashboard and blockchain audit integration

05 Benchmark against simple baselines

Contain threats intelligently. Preserve essential workflows.

ONLINE DEMONSTRATION & PROTOTYPE LINK:
https://github.com/saanviinegii/SafeContain-AI